

# Computer Architecture and Organization

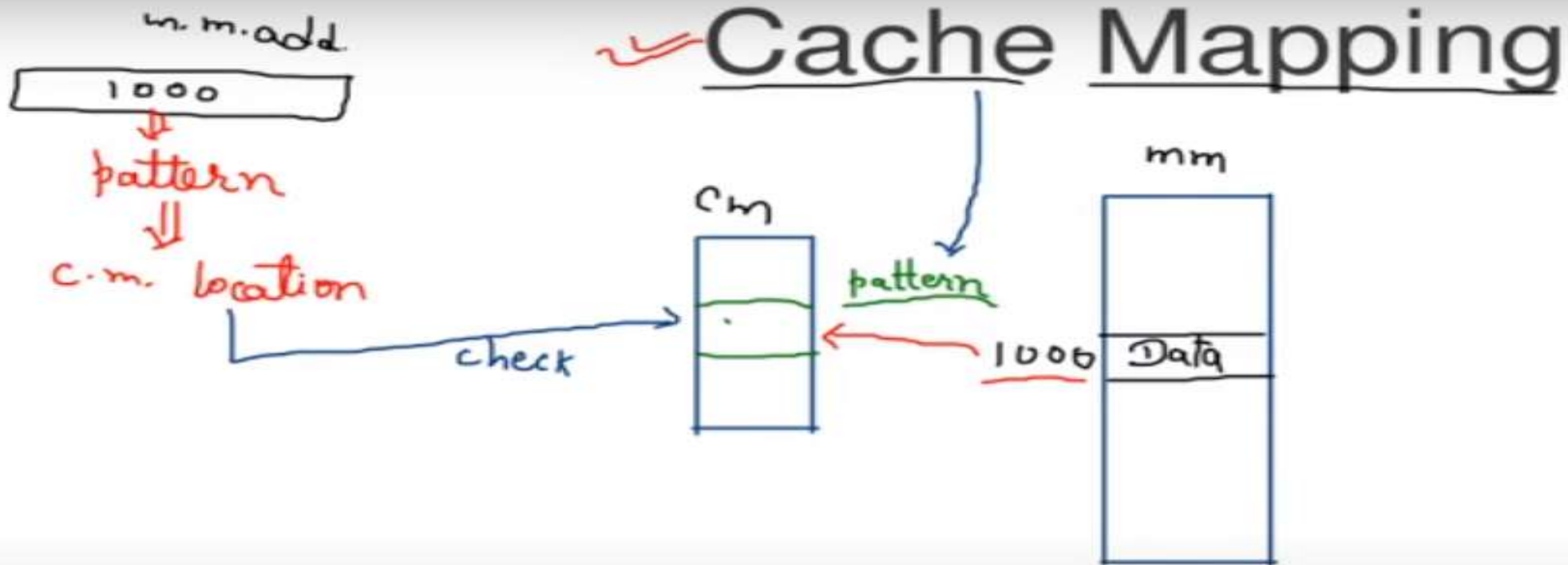
Week\_9

Memory-2

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# Cache Mapping

- CPU always generates the address of main memory.



There is a pattern using which mm content from a specific location is copied in cm. at specific location.

# Cache Memory

- The cache contains a copy of portions of main memory. When the processor attempts to read a word of memory, a check is made to determine if the word is in the cache. If so, the word is delivered to the processor. If not, a block of main memory, consisting of some fixed number of words, is read into the cache and then the word is delivered to the processor.
- Because of the phenomenon of locality of reference, when a block of data is fetched into the cache to satisfy a single memory reference, it is likely that there will be future references to that same memory location or to other words in the block.

# Mapping Function

- Because there are fewer cache lines than main memory blocks, an algorithm is needed for mapping main memory blocks into cache lines
- Three techniques can be used:

## Direct

- The simplest technique
- Maps each block of main memory into only one possible cache line

## Associative

- Permits each main memory block to be loaded into any line of the cache
- The cache control logic interprets a memory address simply as a Tag and a Word field
- To determine whether a block is in the cache, the cache control logic must simultaneously examine every line's Tag for a match

## Set Associative

- A compromise that exhibits the strengths of both the direct and associative approaches while reducing their disadvantages

# Direct Mapping: Pros and Cons

- The direct mapping technique is simple and inexpensive to implement. Its main disadvantage is that there is a fixed cache location for any given block. Thus, if a program happens to reference words repeatedly from two different blocks that map into the same line, then the blocks will be continually swapped in the cache, and the hit ratio will be low (a phenomenon known as *thrashing*).

# Direct Mapping

|                | +  |    |    |    |    |    |    |    |    |    |
|----------------|----|----|----|----|----|----|----|----|----|----|
| Cache          | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| Main<br>Memory | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 |
|                | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
|                | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 |
|                | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 |
|                | :  | :  | :  | :  | :  | :  | :  | :  | :  | :  |
|                | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 |

# Direct Mapping

- Blocks in cache = 10 (0-9)
- Blocks in Main memory = 100 (00-99)



cm. block no. = (mm block no.) mod no. of blocks in cache.

# Direct Mapping

- The mapping is expressed as
- $i = j \bmod m$

where

- $i = \text{cache line number}$
- $j = \text{main memory block number}$
- $m = \text{number of lines in the cache}$

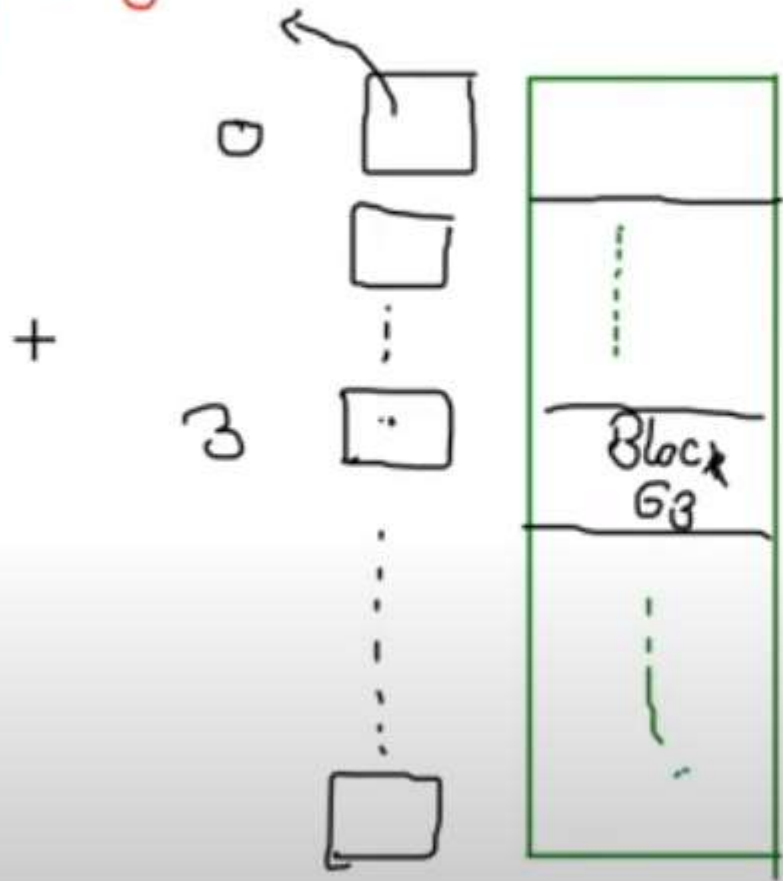


Assume initially cache is empty

# Direct Mapping

| CPU Request (MM block) | Mapping(CM block no.)        | Hit / Miss | Comments                                                           |
|------------------------|------------------------------|------------|--------------------------------------------------------------------|
| Block no.<br>63        | $63 \div 10 = \underline{3}$ | Miss       | Bring mm block no. 63 in cache at block no. = 3                    |
| Block no.<br>93<br>+   | $93 \div 10 = 3$             | Miss       | Bring mm block no. 93 in cache at block no. 3 by replace block 63. |

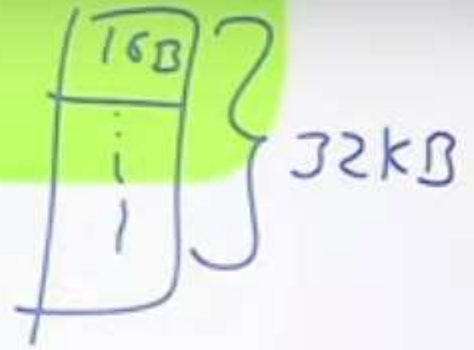
Tag  $\Leftarrow$  { identification }  $\rightarrow$  in identifying which m.m. block is currently present in cache.



# Direct Mapping

| CPU Request<br>(MM block) | Mapping(CM<br>block no.) | Tag      | Hit / Miss | Comments                                                    |
|---------------------------|--------------------------|----------|------------|-------------------------------------------------------------|
| 63                        | $63 \% 10 = 3$           | 6        | Miss       | Bring block 63 from mm to cache at block no. 3, with tag 6. |
| 93                        | $93 \% 10 = 3$           | <u>9</u> | Miss       | Bring block 93 from mm to cache at block 3 by 9.            |

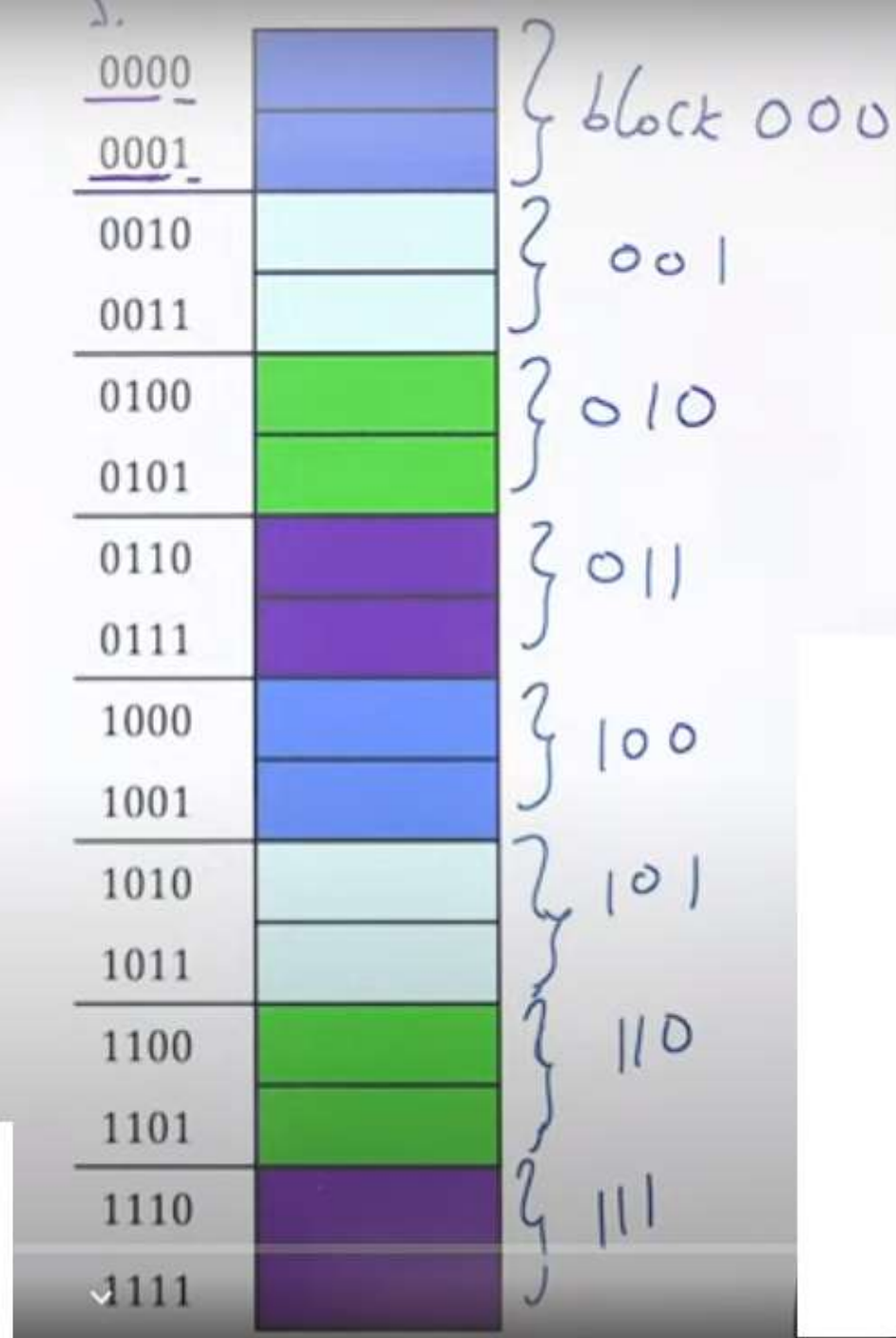
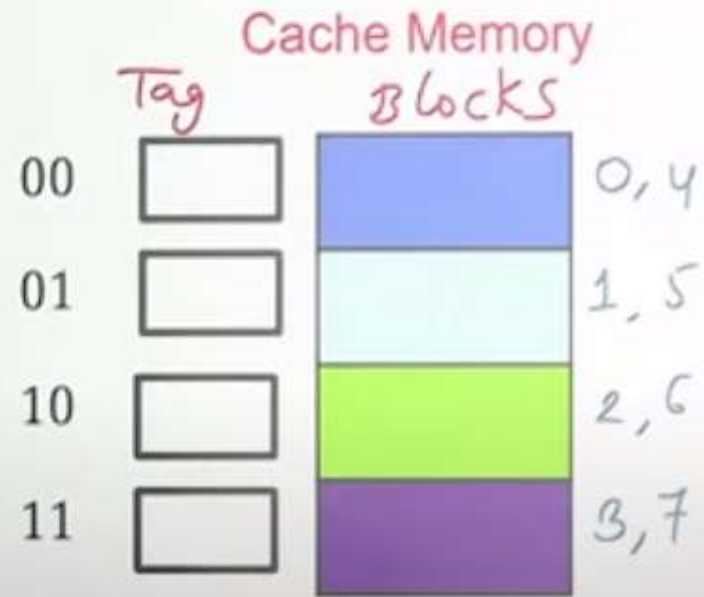
# Direct Mapping



- X Blocks in cache = 4 (00-11)
- X Blocks in Main memory = 8 (000-111)
- X Block Size = 2 Bytes

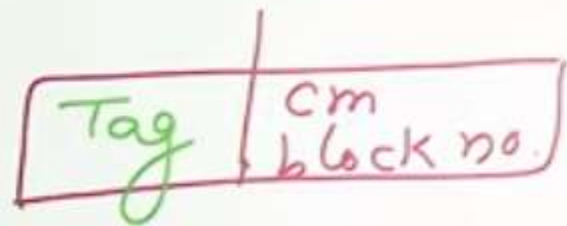
- X Size of Cache memory =  $4 * 2B = 8B$
- X Size of Main memory =  $8 * 2B = 16B = 2^4 B$
- X Size of Main memory address = 4-bits

# Direct Mapping



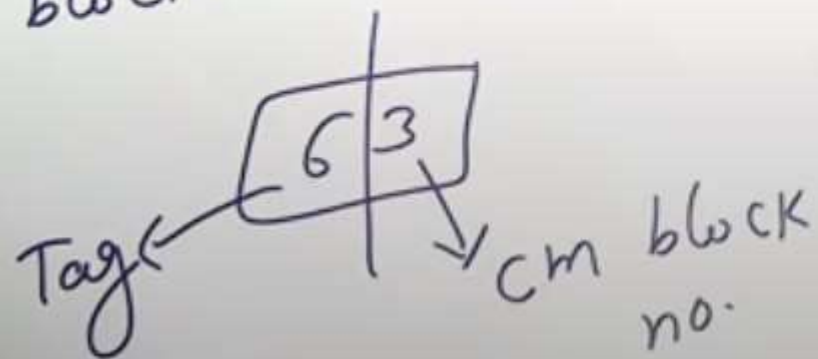
# Direct Mapping

M.M. block no.

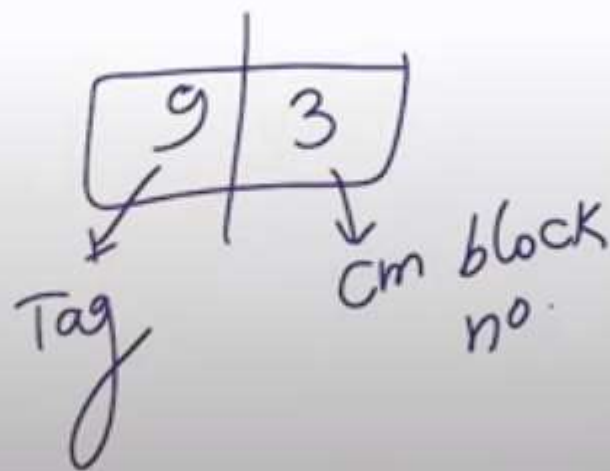


Tag identifies that which block of mm

CPU Requested mm  
block no. 63

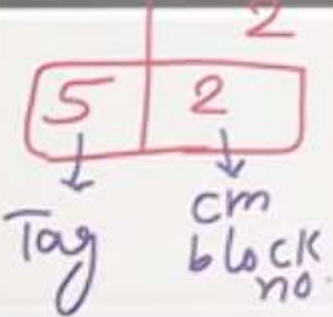
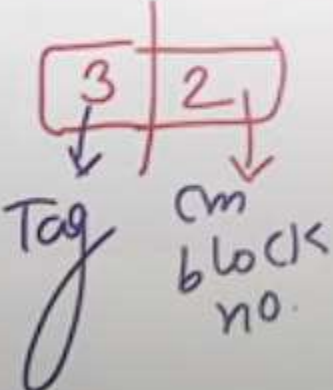


mm block  
no. = 93





# Direct Mapping

| CPU Request (MM block) | Mappin (CM block no.)                                                              | Tag | Hit / Miss | Comments                                                                              |
|------------------------|------------------------------------------------------------------------------------|-----|------------|---------------------------------------------------------------------------------------|
| 52                     |   | 5   | Miss       | Bring block no. 52 of mm into cache at block no. 2, & update tag as 5.                |
| 32                     |  | 3   | Miss       | Bring block no. 32 of mm into cache at block no. 2, by replacing Block 52; and update |

# Direct Mapping

M.M. add.

|                 |             |
|-----------------|-------------|
| MM block<br>no. | Byte<br>no. |
|-----------------|-------------|

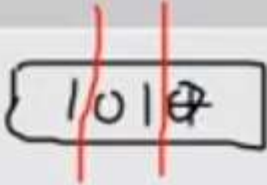
|     |                    |             |
|-----|--------------------|-------------|
| Tag | CM<br>block<br>no. | Byte<br>no. |
|-----|--------------------|-------------|

No. of bits in byte no. =  $\log_2(\text{block size})$

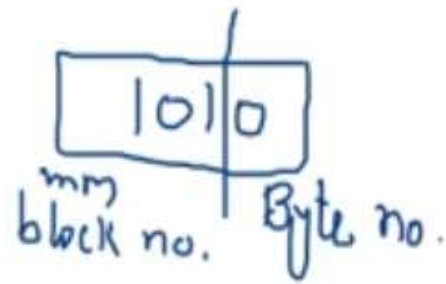
No. of bits in CM block no. =  $\log_2(\text{no. of blocks in cache})$



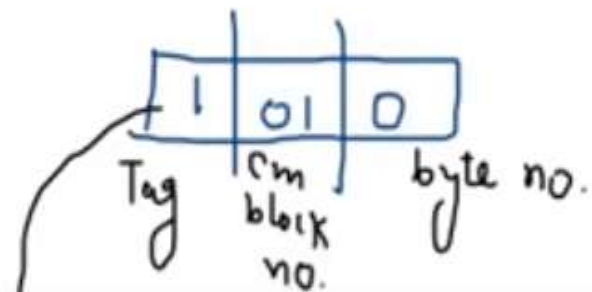
# Direct Mapping

| CPU Request<br>(MM add.) | Mapping(CM<br>block no.)                                                          | Hit /<br>Miss | Comments                                            |
|--------------------------|-----------------------------------------------------------------------------------|---------------|-----------------------------------------------------|
| 1011                     |  | miss          | Bring block 5 from mm to cm at block 1, with tag 1. |
| 1010                     |  | hit           |                                                     |

CPU reg:- mm add 1010



→ Send byte 0 of this block to C  
hit



goto cm block no. = 01

↓  
Tag there = 1

Tag in CPU's reg = 1

match

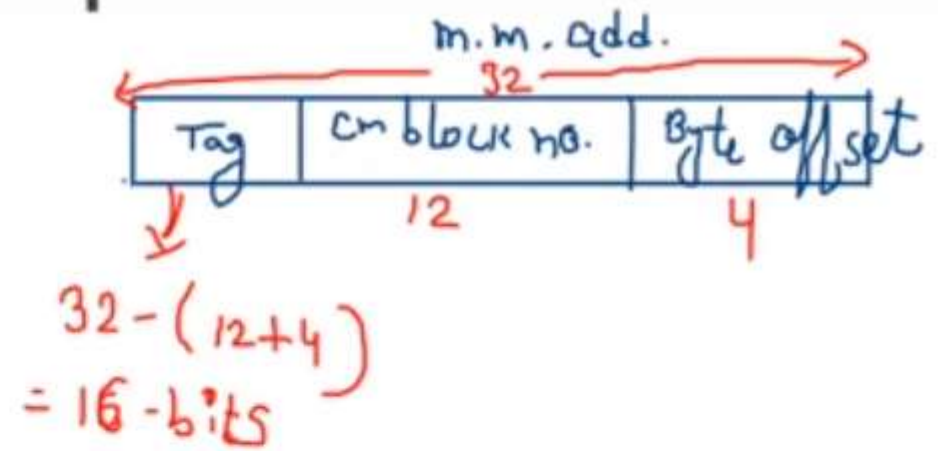
# Direct Mapping Example Question

- Block Size = 16 bytes +
- Size of Cache memory = 64KB
- Size of Main memory address = 32-bits

Questions:

1. Bits in byte offset?
2. Bits in cache block number?
3. Bits in tag?
4. Tag Directory size?

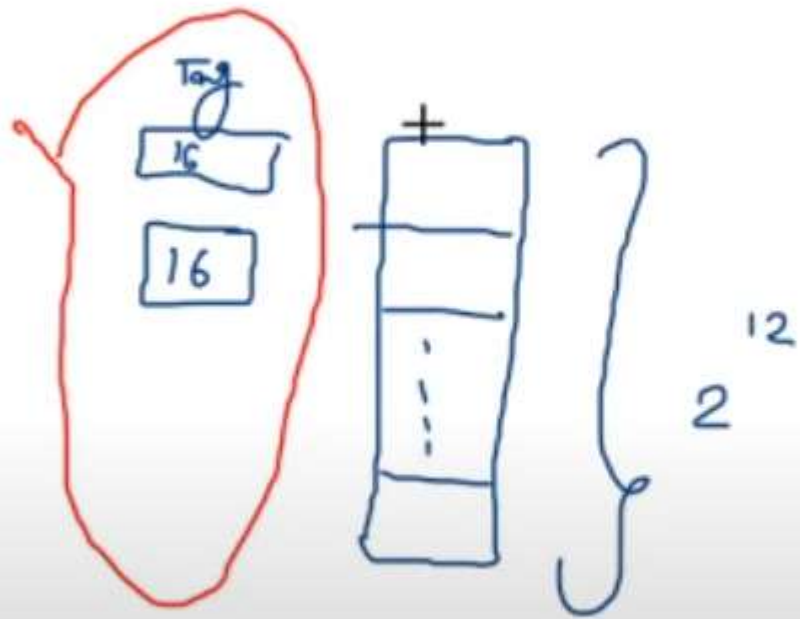
- Block Size  $2^4 \text{B} \leftarrow \boxed{= 16 \text{ bytes}}$
- Size of Cache memory = 64KB
- Size of Main memory address = 32-bits



Questions:

1. Bits in byte offset? = 4-bits
2. Bits in cache block number? = 12-bits
3. Bits in tag? = 16-bits
4. Tag Directory size?

$$\text{no. of blocks in cache} = \frac{\text{cache size}}{\text{block size}} = \frac{64KB}{16B} = 4K = 2^{12} \Rightarrow \text{cm block no} = 12\text{-bits}$$



$$\text{Tag directory size} = \text{no. of blocks in cache} * \text{Tag-bits}$$

$$= 2^{12} * 16\text{-bits}$$

$$= 2^{12} * 2B$$

$$= 8KB$$

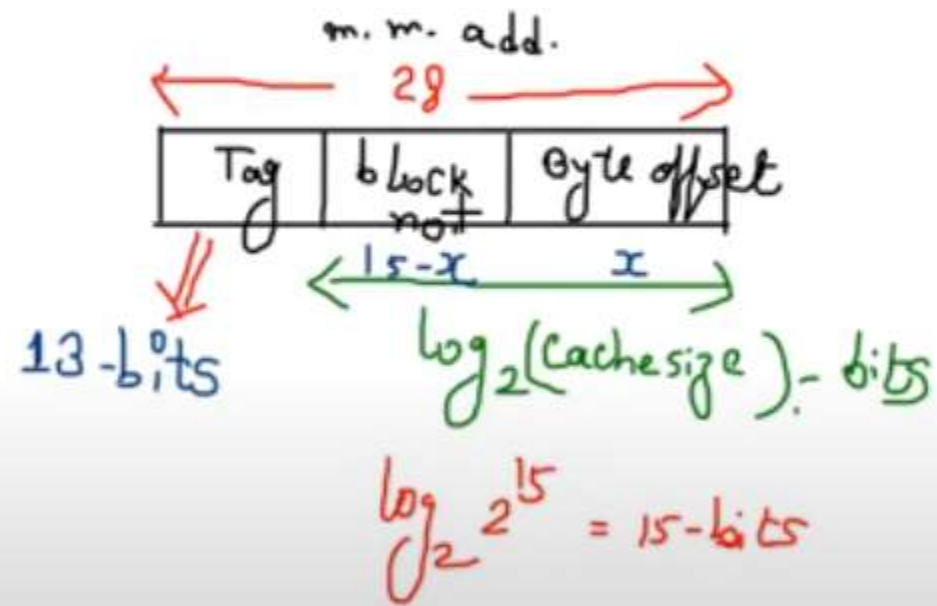
or

$$64K\text{ bits}$$

Q:1

Consider a direct mapped cache of size 32KB. The CPU generates 28 bit addresses. The number of tag bits in main memory address are?

Consider a direct mapped cache of size  $2^{15}$  32KB. The CPU generates 28 bit addresses. The number of tag bits in main memory address are?



$$2^x \text{ block size}$$

$$\frac{2^{15}}{2^x} = 2^{15-x}$$



32-bit architecture CPU

Main Memory Size = 4GB

Cache Size = 512KB

Block Size = 16 Words

No. of bits required for Byte Offset = ?

No of bits required for main memory address = ?

No of bits required for main memory block no. = ?

Index-bits = ?

Tag-bits = ?

Size of Tag Directory = ?



## Direct Mapping

32-bit architecture CPU  $\Rightarrow$  1 word = 32-bits = 4B

Main Memory Size = 4GB =  $2^{32}$ B  $\Rightarrow$  mm add. = 32-bits

Cache Size = 512KB =  $2^{19}$ B

Block Size = 16 Words =  $16 * 4B = 64B$

No. of bits required for Byte Offset = ? 6-bits

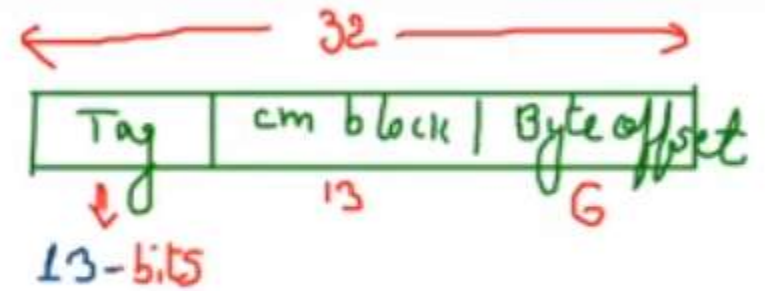
No of bits required for main memory address = ? 32-bits

No of bits required for main memory block no. = ? 26-bits

Index-bits = ? 13-bits

Tag-bits = ? 13

Size of Tag Directory = ?  
 $\text{No. of blocks in cache} * \text{Tag bits}$   
 $= 2^{13} * 13\text{-bits}$   
 $= 104k \text{ bits}$

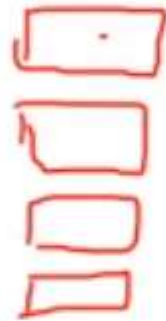


$$\text{No. of blocks in cache} = \frac{2^{19}}{2^6} = 2^{13}$$

# Practice Question 11

Consider a 64 bytes direct mapped cache with a block size of 16 bytes. Main memory size is 256bytes. Currently in the cache, the blocks are having tags as follows:

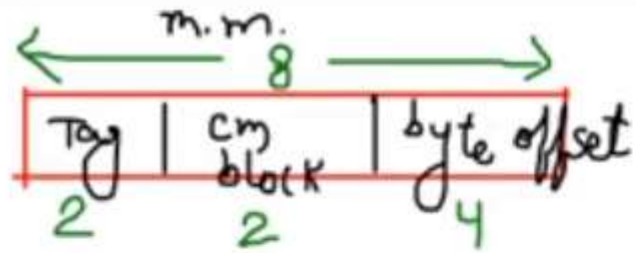
| Block | Tag |
|-------|-----|
| ✓ 00  | 10  |
| 01    | 10  |
| 10    | 00  |
| 11    | 01  |



Identify the correct statement with respect to the availability of the main memory data into cache? +

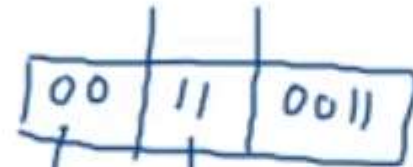
- (A) Main memory byte number 51 and 134 are present in cache
- (B) Main memory byte number 156 and 132 are present in cache
- (C) Main memory byte number 156 and 178 are present in cache
- (D) Main memory byte number 33 and 51 are present in cache

## Solution



+

$$(i) (51)_{10} = (00110011)_2$$

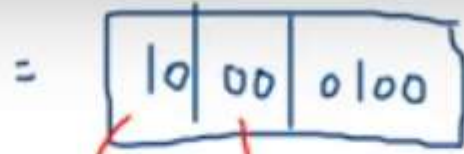


block 3

↓

in cache tag = 01  
tag = 00 → no match  
miss

$$\underline{(132)}_{10} = (10000100)_2$$



block = 0



goto 0 in cache

Tag = 10

Tag = 10

} → match

!!  
hit

Consider a 64 bytes direct mapped cache with a block size of 16 bytes. Main memory size is 256bytes. Currently in the cache, the blocks are having tags as follows:

$2^8$  ↓

| Block | Tag |
|-------|-----|
| 00    | 10  |
| 01    | 10  |
| 10    | 00  |
| 11    | 01  |



no of block =  $\frac{64}{16} = 4$

Identify the correct statement with respect to the availability of the main memory data into cache?

- ~~(A)~~ Main memory byte number 51 and 134 are present in cache
- ~~(B)~~ Main memory byte number 156 and 132 are present in cache
- (C) Main memory byte number 156 and 178 are present in cache
- ~~(D)~~ Main memory byte number 33 and 51 are present in cache



Direct Mapping Question: Assume a computer has 32 bit addresses. Each block stores 32 bytes. A direct-mapped cache has 256 blocks. In which block (line) of the cache would we look for each of the following addresses?

1. 1A2BC012
2. FFFF00FF
3. 12345678
4. C109D532

Direct Mapping Question: Assume a computer has 32 bit addresses. Each block stores 32 bytes. A direct-mapped cache has 256 blocks. In which block (line) of the cache would we look for each of the following addresses?

1. 1A2BC012 → 1100 0000 0000 0000 → 0
2. FFFF00FF → 0000 0000 1111 1111 → 7
3. 12345678 → 0000 0000 1111 1111 → 7
4. C109D532 → 1101 0101 1001 1111 → 179

